

MINI CLASS ASSESSMENT GUIDE

Course: ASP.NET Core MVC

Duration: 2 Hours

Title: Enhancing an ASP.NET Core MVC App with APIs and CDNs

1. Assessment Overview

This mini assessment is designed to evaluate students' ability to integrate **external APIs** and use **Content Delivery Networks (CDNs)** within an ASP.NET Core MVC application.

Students will be provided with (or expected to quickly scaffold) a basic MVC application and enhance it by:

- Consuming an external API
 - Displaying API data in the UI
 - Integrating frontend libraries via CDN
 - Using asynchronous requests (AJAX / Fetch API)
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2. Scenario

You are developing a simple **Event Booking Web Application**. The basic structure already allows users to view venues.

Your task is to enhance the application by:

1. Displaying additional dynamic data from an external API
 2. Improving the UI using CDN-hosted libraries
 3. Making at least one asynchronous request to your backend or an external API
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3. Learning Outcomes

By completing this task, students should be able to:

- Consume external APIs using HttpClient
- Deserialize and use JSON data in ASP.NET Core
- Integrate frontend frameworks using CDNs
- Perform asynchronous requests using JavaScript (Fetch/AJAX)
- Display dynamic data in Razor views

4. Task Requirements

4.1 External API Integration (Core Requirement)

Choose ONE of the following API integrations:

Option A: Weather API

- Fetch weather data for a selected venue location
- Display temperature and conditions on the venue details page

Option B: Maps API

- Display a map showing the venue location
- Use latitude/longitude or address

Option C: Public Events API (or similar)

- Fetch and display related events based on venue location

4.2 API Consumption (Backend)

Students must:

- Create a service class (e.g., ApiService)
- Use HttpClient to call the external API
- Handle API responses properly
- Return processed data to the controller

4.3 UI Enhancement Using CDN

Students must include at least ONE CDN-based library:

Examples:

- Bootstrap (for styling)
- jQuery (for DOM manipulation)
- Chart.js (for simple data visualization)

The CDN must be linked directly in the layout or view file.

4.4 Asynchronous Interaction (Frontend)

Students must implement at least ONE asynchronous feature using:

- Fetch API OR
- jQuery AJAX

Examples:

- Load weather data when a button is clicked
 - Dynamically update venue information without reloading the page
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5. Minimum Implementation Guide

Students should aim to complete the following:

1. Create or use an existing MVC project
 2. Add a new service for API calls
 3. Call the API from a controller
 4. Pass data to a view
 5. Display results clearly
 6. Add CDN link(s) to improve UI
 7. Implement one asynchronous request
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6. Assessment Criteria

Criterion	Weight
External API Integration (working and correct)	40%
Use of HttpClient and data handling	20%
CDN Integration and UI improvement	15%
Asynchronous Feature (Fetch/AJAX)	15%
Code Structure and Clarity	10%

7. Deliverables

Students must submit:

- Source code (zipped or repository link)
 - Brief README including:
 - API used
 - Description of implemented feature
 - Any setup instructions
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8. Constraints

- Time limit: 2 hours
 - Focus on functionality over design perfection
 - Use of external documentation is allowed
 - API keys (if required) must be handled securely
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9. Notes for Students

- Start simple and ensure the API call works before improving UI
 - If an API fails, mock or simulate the response to demonstrate understanding
 - Prioritize completing all required sections before attempting enhancements
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End of Mini Assessment Guide